Sander Lanbo Shi

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WORK EXPERIENCE

SAMPURA RESEARCH | Founding Member of Technical Staff — Seattle, WA | Feb 2026 - Present

- Co-first author of *Towards Human-AI Complementarity for AI Safety* (arXiv preprint, under review at **ICLR 2027**): the largest known suite of safety-relevant complementarity tasks (**3,957** items, 8 datasets) with **35,613** judgements from 416 human raters. Joined via the SPAR & MARS fellowship; hired as founding technical staff.
- Led all engineering for the paper: re-architected the experiment pipeline (ingestion, splits, inference, confidence calibration, hybridisation) that produces the leaderboard across **260** routing configurations.
- Proposed and led the oracle-trained router: a question-only hidden-state probe on **Qwen3.5-9B** (PCA + ridge) trained on paired human/AI correctness to predict when to route to a human; integrated probe, Confidence **LoRA**, and **SWAG-LoRA** signals into one compare-confidence evaluation.
- Built the human rating platform used for all **Prolific** data collection, including the Top-N and iterative subtask delegation (ISD) assistance interfaces; co-led data collection and produced the paper's figures and tables.

SUNBURST (STARTUP) | AI Engineer — Seattle, WA | Nov 2025 - Jul 2026

- Engineered a multi-agent orchestration harness for an AI web-design studio, generating, diffing, merging, and self-healing site code, with **Braintrust** evals to select the optimal model per task.
- Designed and built an agent memory system, persisting design context and prior edits to improve iterative edit quality.

GOOGLE | Software Engineer — Seattle, WA | Jan 2022 - Oct 2025

- Led efforts to ship auto budget allocation for DV360, adopted by **50%+** of advertisers with **20%+** performance gains.
- Built a publisher deal-spend tracking system (**$3B+**, **100+** advertisers) with a caching layer serving near-real-time aggregates, cutting query latency from **minutes to under a second** at **30-minute** freshness.
- Ran two zero-downtime migrations (schema redesign, int32-to-int64 ID widening) via dual-write, backfill, and sampled-checksum reconciliation, guaranteeing no data loss.
- Built API access letting partners upload custom ML bidding algorithms, adopted by major enterprise advertisers.

HULU (DISNEY) | Software Engineer — Seattle, WA | Jul 2019 - Jan 2022

- Built a custom WebKit-based cross-platform browser engine (C++, Java) for Linux, Windows, macOS, and UWP, shipping to **7M+** devices from Fire TV to Xbox One.
- Designed and implemented a memory tagging system and pressure handler, cutting resource usage **20%** on low-powered devices.

EDUCATION

CARNEGIE MELLON UNIVERSITY | B.S. Computer Science, Minor in Mathematics — Class of 2019

- **Relevant courses:** ML, NLP, AI, Distributed Systems, Complexity Theory, Computer Vision, Discrete Math
- **Other:** Dean's List, Teaching Assistant for Dr. Manuel Blum in Complexity Theory

VOLUNTEERING

CSEdWeek @ Google (2022-2024): Taught AI and software engineering to grades 6-12.
Ada Developers Academy (2019-2024): Taught CS and provided interview coaching to gender-diverse communities.

SKILLS

Programming languages | Python, C++, Java, JavaScript / TypeScript
Technologies | PyTorch, Braintrust, scikit-learn, Linux (avid Arch Linux user), React
Music | Multi-instrumentalist, Music Theory / Jazz nerd. Playing for 5 bands in Seattle.
Languages | English (native / bilingual), Mandarin (native / bilingual), French (elementary and improving...)